

JAPNIT SINGH SAWHNEY

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EDUCATION

Thapar Institute Of Engineering & Technology, Patiala	Aug 2023 - Present
B.Tech in Computer Engineering	CGPA: 9.10
St. Mary's Academy, Meerut	Apr 2022 - Mar 2023
Senior Secondary School certificate	Percentage: 94.5%
St. Mary's Academy, Meerut	Apr 2020 - Mar 2021
Senior Secondary School certificate	Percentage: 95.8%

PROJECTS

UniVerse - CampusGPT Assistant for TIET Students



- Built a hybrid AI-powered campus assistant for Thapar University that answers real-time student queries on timetables, faculty, subjects, cafeteria menus, certificates, DoAA procedures, dispensary info, and campus navigation, using LLM-based intent detection integrated with structured (MySQL) and JSON-backed services.
- Implemented an interactive campus navigation system using weighted graphs, Dijkstra's shortest-path algorithm, Leaflet/OpenStreetMap for visualization, OSRM for routing, and walking time estimation.
- **Tech Stack** - MySQL, Google Gemini API, Vercel, Render, Python

AI Interview Coach



- Built an AI-powered interview preparation platform that uses Retrieval-Augmented Generation (RAG) to generate personalized interview questions from resumes and job descriptions, conduct multi-round mock interviews, and provide automated performance evaluation.
- Implemented semantic search and personalized question generation using ChromaDB, LangChain, HuggingFace embeddings, and Groq LLMs, with AI-driven answer evaluation and interview reporting.
- **Tech Stack** - Python, LangChain, ChromaDB, Groq API, HuggingFace Embeddings, RAG

Bangalore Traffic Prediction



- Built a Bangalore traffic congestion prediction system using historical traffic, weather, environmental impact, and temporal features, deployed via a FastAPI backend to deliver real-time congestion levels, categories, and analytical insights.
- Implemented a stacked Random Forest-XGBoost regression model with spatial and temporal feature engineering, achieving RMSE = 2.031 and $R^2 = 0.995$, outperforming baseline models and results reported in related research studies.
- **Tech Stack** - Python, FastAPI, Google Gemini API, HTML, CSS

TECHNICAL SKILLS

Languages: Python, C, C++, Java

Libraries & Frameworks: LangChain, Hugging Face Transformers, NumPy, Pandas, Scikit-learn, FastAPI

Artificial Intelligence/Machine Learning: Generative AI, Hugging Face, Retrieval Augmented Generation

Cybersecurity & Network Analysis Tools: Wireshark, LordPe, Ollydbg, Kali Linux, Cisco Packet Tracer

Tools & Technologies: MySQL, Chroma, HTML/CSS, Ollama, Vercel, Linux, VS Code, Render

Core CS Concepts: Object-Oriented Programming (OOP), Operating Systems (OS), DBMS, Data Structures and Algorithms, Computer Networks, Artificial Intelligence, Cybersecurity & Network Protocols

ACHIEVEMENTS

Gold medalist in 3600m cycle race.

Silver medalist in Skating Hockey (States)

CERTIFICATION

EC-Council Cybersecurity Workshop (1 CPE Credit)

Software Engineer Intern (HackerRank)